

Lecture 06. Existence and Uniqueness for Systems

SIR Model

Phase Plane

— end of chapter 2.

General announcements:

- Midterm: 6pm - 7:20pm + 10 min to scan.
 - Write complete solutions, i.e., explain how you got the result.
 - * Show your ID in ~~all~~ your scans* 
 - Chapter 1+2 of the book (p. 1 - 237) + section 3.6., 7.1.
 - * 8 questions (10 min/question) + 1 extra
- Note that 10 points = 1% of final grade midterm: 200pt.
- Weekly assignments.

- HW #1: 30 pt	Lab 1: 15pt	
HW #2: 30 pt	Lab 2: 20pt	
easy → HW #3: 10 pt	Lab 3: 15pt	*5 = 20pt
HW #4: 20 pt	Lab 4: 20pt	*5 = 25pt
HW #5: 10 pt	Lab 5: 15pt	*5 = 20pt
100pt	Lab 6: 15pt	100pt.

Fri, Sun 17th: no class. W - 3:40 - 5:40pm office hours.

2.5. Euler's Method Systems (Matlab 4)

Suppose we have an IVP (different notations, same thing)

$$\frac{d\vec{p}}{dt} = \vec{V}(t, \vec{p}), \vec{p}(t_0) = \vec{p}_0 \quad \text{or} \quad \frac{d\vec{Y}}{dt} = \vec{F}(t, \vec{Y}), \vec{Y}(t_0) = \vec{Y}_0$$

Remember:

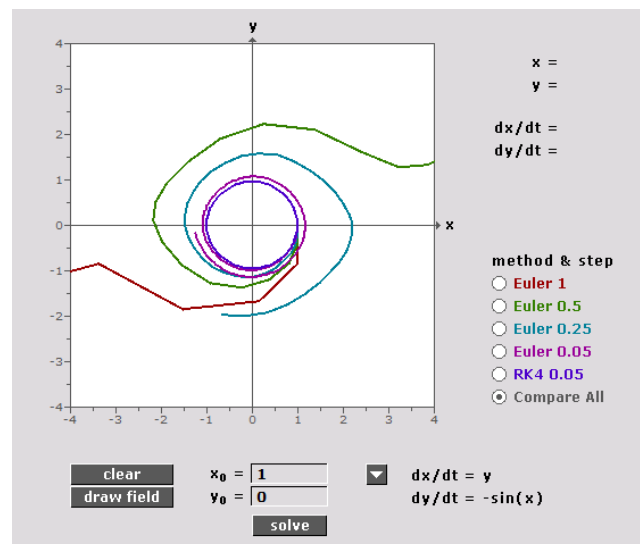
$$y(t_1) = y(t_0) + \Delta t f(t_0, p(t_0))$$

new version:

$$\vec{Y}(t_{k+1}) = \vec{Y}(t_k) + \Delta t \cdot \vec{F}(t_k, \vec{Y}(t_k))$$

Example: Compare Euler with step-size $\Delta t = 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{20}$ with Runge-Kutta $\Delta t = \frac{1}{20}$ on the pendulum (preclass)

$$\begin{cases} \dot{x} = y \\ \dot{y} = -\sin(x) \end{cases} \quad \begin{matrix} x(0) = 1 \\ y(0) = 0 \end{matrix}$$



2.6. Existence and Uniqueness for systems

theorem: Consider the IVP

$$\frac{d\vec{Y}}{dt} = \vec{F}(t, \vec{Y}), \quad \vec{Y}(t_0) = \vec{Y}_0, \quad t_0 \in \mathbb{R}, \vec{Y}_0 \in \mathbb{R}^n$$

If $\vec{F}$ is continuously differentiable, then there is $\varepsilon > 0$ and $\vec{Y}: (t_0 - \varepsilon, t_0 + \varepsilon) \rightarrow \mathbb{R}^n$ such that $\vec{Y}$ satisfies the IVP and is unique.

$$\begin{aligned} x' &= f(t, x, y) \\ y' &= g(t, x, y) \end{aligned}$$

$\frac{\partial f}{\partial t}, \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial g}{\partial t}, \frac{\partial g}{\partial x}, \frac{\partial g}{\partial y}$ all exist and be continuous.

Natural questions.

1. Why $\varepsilon > 0$? Finite time blow-up.

$$\text{Ex } \begin{cases} x' = x^2 + 1 \\ y' = 1 \end{cases} \rightarrow (x, y) = (\tan(t), t)$$

2. What does uniqueness mean?

A: Two different solutions can't start at the same place and same time.

Remarks when drawing stuff.

proof on p.207.

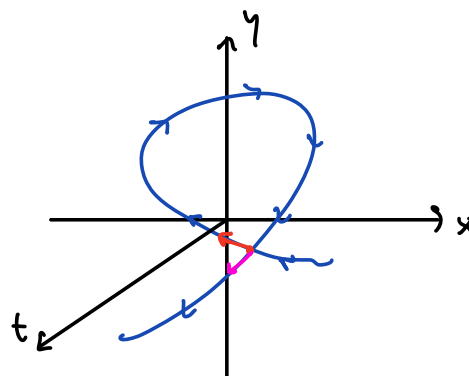
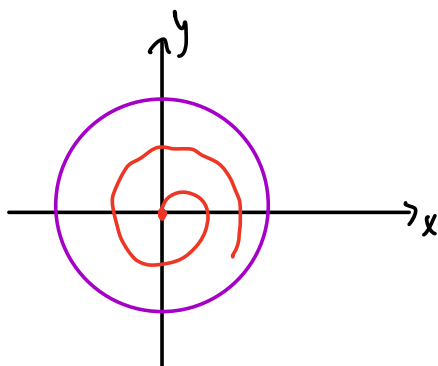
Autonomous systems.

vs

Non-Autonomous systems

$$\vec{Y}' = F(\vec{Y})$$

$$\vec{Y}' = F(t, Y)$$



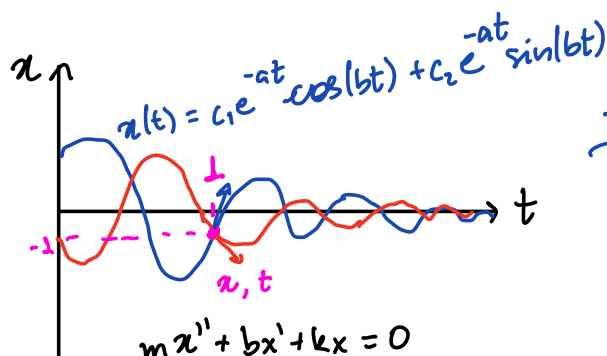
Suppose underdamped. (matlab 3)

2nd order linear eq.

$$m\ddot{x} + b\dot{x} + kx = 0 \quad \text{vs}$$

System of 1st order.

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= -\frac{k}{m}x - \frac{b}{m}y \end{aligned} \quad \left\{ \begin{array}{l} \text{autonomous.} \end{array} \right.$$

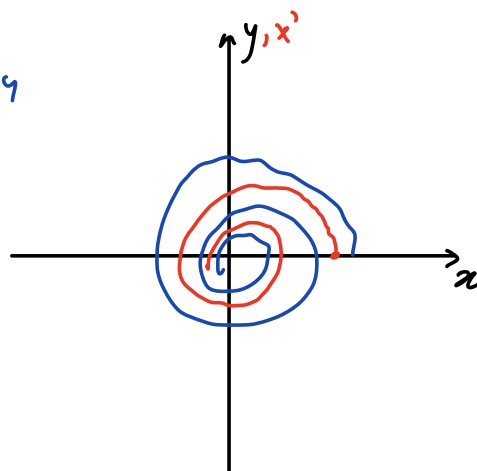


not precisely
the same
solution.

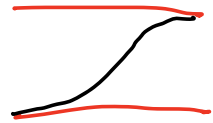
$$m\ddot{x} + b\dot{x} + kx = 0$$

$$x(1) = x'(1) = -1$$

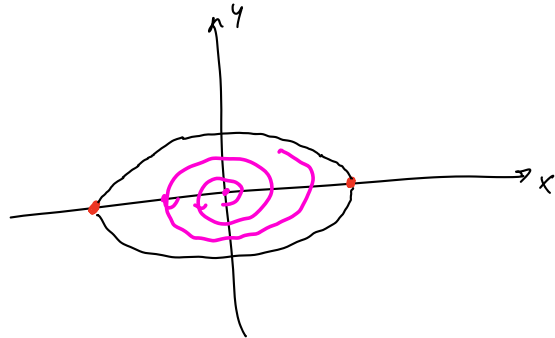
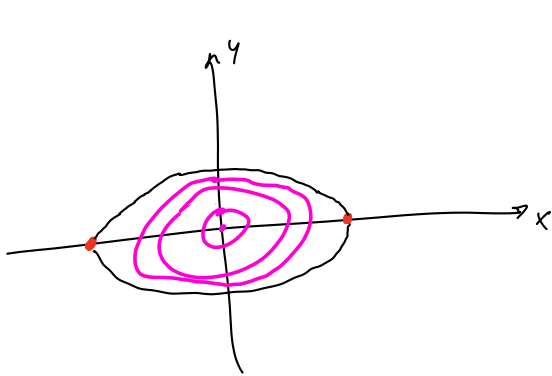
$$x'(1) = 2 \quad x''(1) = -2$$



Conclusion for 2D autonomous systems.



(Jordan Curve)

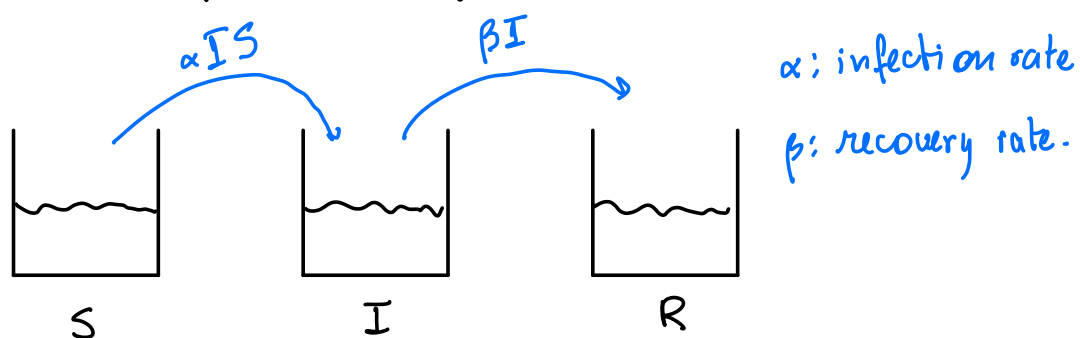


2.7. the SIR model of an epidemic (before covid)

$S(t)$: fraction of susceptible people at time t

$I(t)$: fraction of infected people at time t

$R(t)$: fraction of recovered people at time t



Remark.

$$S + I + R = 1 \Rightarrow R = 1 - S - I.$$

so only study

$$S' = -\alpha SI$$

$$I' = \alpha SI - \beta I$$

Question:

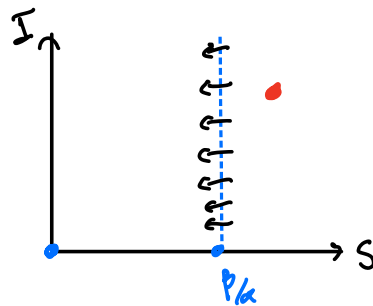
1. Constant solutions?

$$S' = 0 \Rightarrow S = 0 \text{ or } I = 0.$$

$$I' = 0 \Rightarrow I = 0 \text{ or } S = \beta/\alpha.$$

$$(0, 0), (\beta/\alpha, 0)$$

2. Phase plane analysis. $S = \beta/\alpha$.



$$-\alpha \cdot \frac{\beta}{\alpha} \cdot I = -\beta I$$

3. Level sets.

$$\begin{aligned} \frac{dI}{dS} &= \frac{\alpha SI - \beta I}{-\alpha SI} = \frac{\alpha SI}{-\alpha SI} + \frac{\beta I}{\alpha SI} \\ &= -1 + \frac{\beta}{\alpha} \frac{1}{S} \end{aligned}$$

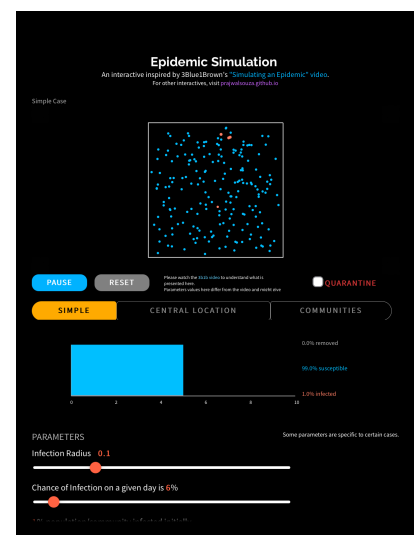
$$\int dI = \int \left(-1 + \frac{\beta}{\alpha} \frac{1}{S}\right) dS$$

$$H(I, S) = I + S - \frac{\beta}{\alpha} \ln(S)$$

$$I = -S + \frac{\beta}{\alpha} \ln S + c.$$

Fun exercise: show that $S+I+R=1$.

<https://prajwalsouza.github.io/Experiments/Epidemic-Simulation.html>

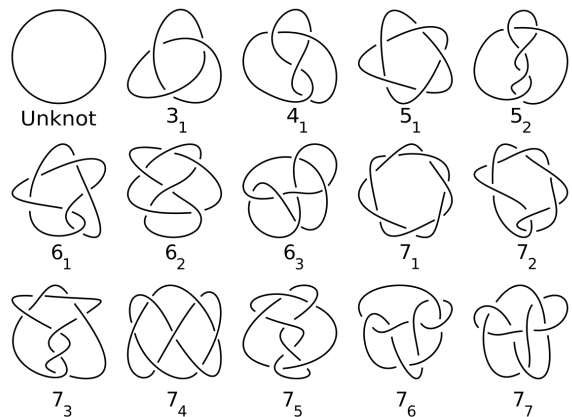
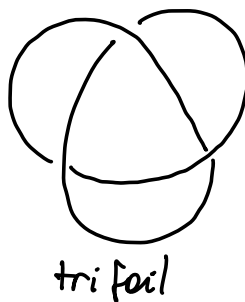
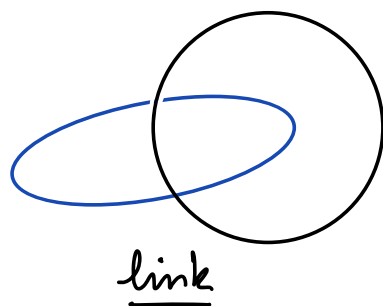


2.8. the Lorenz Equations.

Natural question: geometry of 3D autonomous systems?

Easy, right? knot really.

Periodic solutions might look like

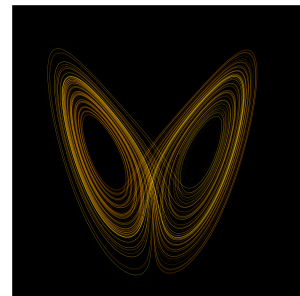


Ex. Lorenz Systems.

$$x' = \sigma(y - x) \quad \sigma = 10$$

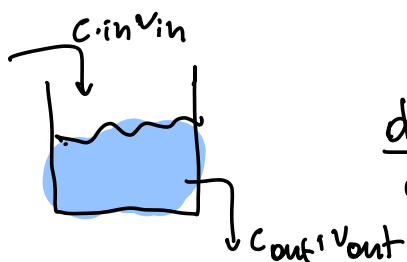
$$y' = \rho x - y - xz \quad \rho = 28$$

$$z' = -\beta z + xy \quad \beta = 8/3.$$



* Small changes in the initial conditions leads to a huge difference in the solutions (i.e. can't trust long term numerical solutions).

"butterfly effect": a distant butterfly flapping its wing could affect you passing this course.



$$\frac{dm(t)}{dt} = c_{in} \cdot v_{in} - c_{out} \cdot v_{out}$$

$$c_{in} = \text{constant} = c.$$

$$v_{in} = \text{const.} = b$$

$$v_{out} = \text{const.} = a$$

$$\begin{cases} v' = b - a \\ v(0) = 10 \end{cases} \quad v(t) = (b-a)t + 10$$

$$b - a \leq 0$$

$$m' = c \cdot b - \frac{m(t)}{v(t)} \cdot a =$$

Quiz 6.

①

$$x' = ax - bxy$$

$$y' = -cy + dxy$$

a) x : prey, y predator

$$b) H(x, y) = a \ln|y| + c \ln|x| - by - dx.$$

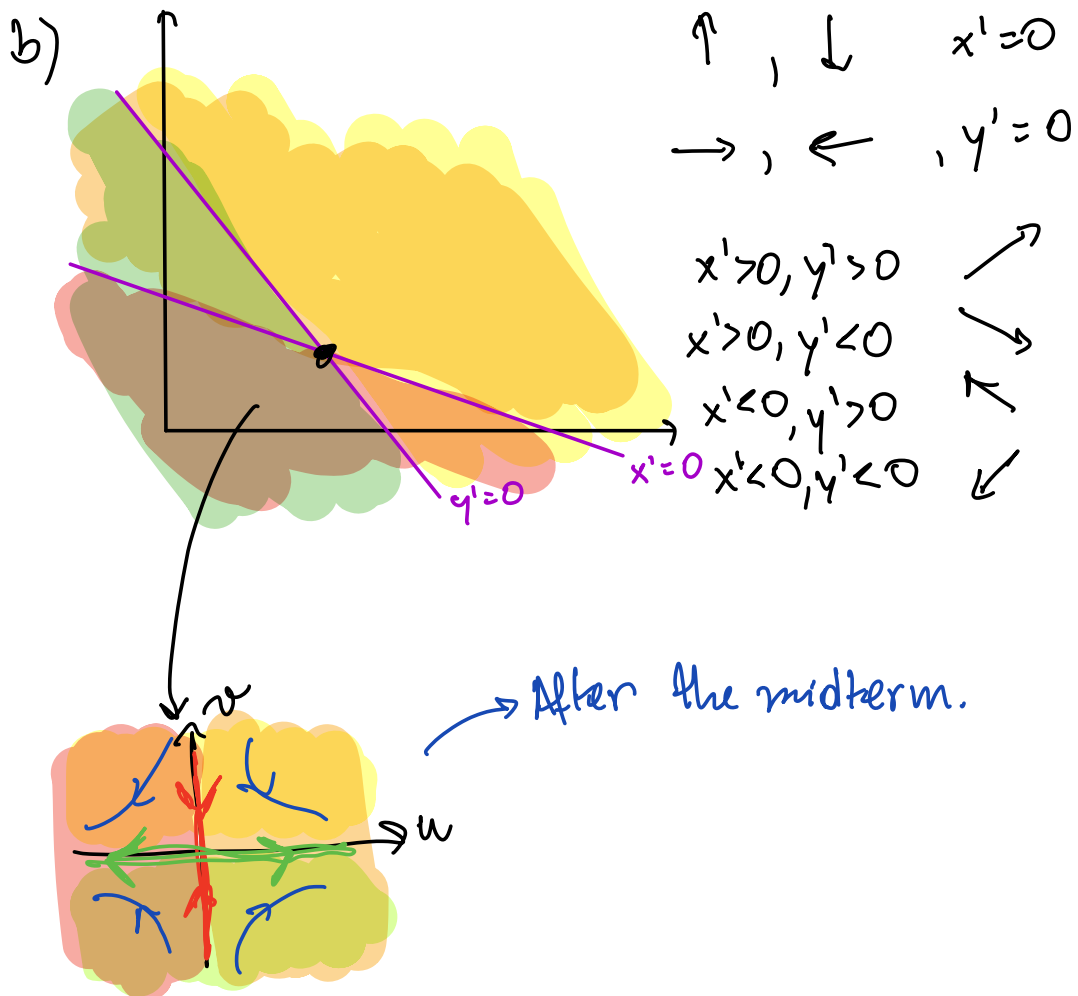
$$\frac{dy}{dx} = \frac{-cy + dxy}{ax - bxy} = \frac{y}{x} \frac{(dx - c)}{(a - by)}$$

$$\int \frac{a - by}{y} dy = \int \frac{dx - c}{x} dx$$

$$a \ln|y| - by = dx - c \ln|x| + k$$

c) solutions oscillate, when prey is low, predator decrease,
which leads to increase in prey
when prey is high, predator increase,
leads to prey low.

2. a) $x' = 0$
 $y' = 0$



2. c (homework)

$y' = y + 2t \sin(3t)$, $y_h = k e^t$, $y_p = ?$

$y_p(t) + 2t \sin(3t) = A s(3t) + B c(3t) + C t s(3t) + D t c(3t) + 2t s(3t)$
 $y_p' = 3A c(3t) - 3B s(3t) + C s(3t) + 3C t c(3t) + D c(3t) - 3D t s(3t)$

$$\begin{cases} 3A + D = B \\ -3B + C = A \\ 3C = D \\ -3D = C + 2. \end{cases} \rightarrow \left[\begin{array}{cccc|c} 3 & -1 & 0 & 1 & 0 \\ -1 & -3 & 1 & 0 & 0 \\ & & 3 & -1 & 0 \\ & & -1 & -3 & 2 \end{array} \right]$$

A
b

A \ b

3. $\alpha y'' + \beta y' + \gamma y = 0$ $\lambda: \alpha \lambda^2 + \beta \lambda + \gamma = 0, \lambda = \frac{-\beta}{2\alpha} \pm \frac{\sqrt{0}}{2\alpha}$

$$y_2(t) = t e^{\lambda t}, \quad y_2' = e^{\lambda t} + \lambda t e^{\lambda t}, \quad y_2'' = 2\lambda e^{\lambda t} + t \lambda^2 e^{\lambda t}$$

$$(f(t) \cdot e^{\lambda t})' = f' \cdot e^{\lambda t} + \lambda f e^{\lambda t} = (f' + \lambda f) e^{\lambda t}$$

$$\alpha y_2'' + \beta y_2' + \gamma y_2 = \alpha(2\lambda e^{\lambda t} + t \lambda^2 e^{\lambda t}) + \beta(e^{\lambda t} + \lambda t e^{\lambda t}) + \gamma t e^{\lambda t}.$$

$$= e^{\lambda t} \left[\underbrace{t(\alpha \lambda^2 + \beta \lambda + \gamma)}_0 + \underbrace{2\lambda \alpha + \beta}_{2\alpha(\frac{-\beta}{2\alpha}) + \beta}_0 \right] = 0.$$

$$\lambda = \frac{-\beta \pm \sqrt{\beta^2 - 4\alpha\gamma}}{2\alpha} = \frac{-\beta}{2\alpha}$$

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8 questions: 200pt { 220pt $\approx$ 22%
 1 extra : 20pt.

$$b. \quad x'' + x' + 2x = 0$$

$$a) \quad \lambda = \frac{-1 \pm \sqrt{1^2 - 4 \cdot 1 \cdot 2}}{2} = \frac{-1 \pm \sqrt{-7}}{2} = -\frac{1}{2} \pm \frac{\sqrt{7}}{2}i$$

$$y(t) = c_1 e^{-\frac{t}{2}} \cos\left(\frac{\sqrt{7}}{2}t\right) + c_2 e^{-\frac{t}{2}} \sin\left(\frac{\sqrt{7}}{2}t\right)$$

$$\omega = \frac{\sqrt{7}}{2} \Rightarrow v = \frac{\omega}{2\pi} = \frac{\sqrt{7}}{4\pi}$$

$$b) \quad v = \frac{\omega}{2\pi} \quad \omega = 2\pi v$$